CCCA Cell-State Analysis — Dong et al. 2020, Prostate Cancer

Prostate adenocarcinoma + small-cell NEPC · 10x Genomics · 9,698 malignant cells (4 patients)

GO / NO-GO GATES

GATE 1	GATE 2	GATE 3
NO-GO	NA	NA

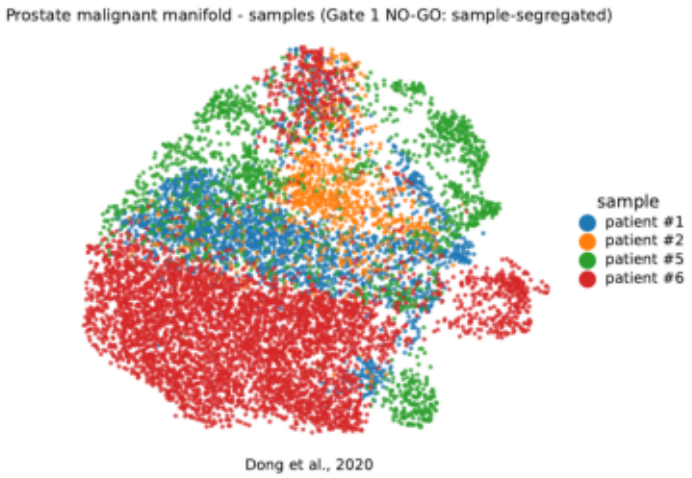
Gate 1 — mixing + continuity: NO-GO

The malignant manifold clusters BY SAMPLE. After Harmony(sample) the kNN graph is a single component and median iLISI rises 1.00→1.31 — the scalars a naive test reads as "integrated". But on the regenerated embedding each patient holds its own territory: self-neighbour enrichment 3.3×/7.3×/3.7×/1.7× (patients #1/#2/#5/#6); 3 of 4 cross the 2.5× segregation line, and patient #6 (small-cell NEPC, ~49% of cells) forms its own pole by size. Integration does not fix it — enrichment barely moves across theta 2–20 and iLISI plateaus at 1.31 (theta=50 collapses). It is not a PC-count artifact: the archipelago holds across n_pcs = 20/30/40 (worst 6.8→7.4×). Every sample carries the same embryonic modules — the geography is inter-sample clonal/CNV + histologic (adeno vs NEPC) heterogeneity, not a shared, sample-integrated cell-state continuum.

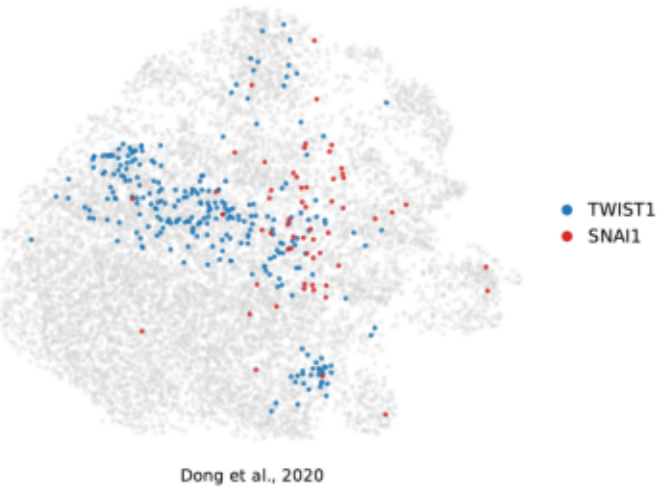
Per protocol (Gate 1, HIGH): do not force a trajectory. Gates 2 & 3 are NA (chain stops).

STEP FIGURES

01 · samples.pdf — sample-segregated manifold



02 · expression.pdf — EMT-TF overlay (TWIST1 · SNAI1)



PARAMETER CHOICES & RATIONALE

Min-cell threshold	floor ≥100 (all 4 samples kept, 915–4765 cells). Gap-finding rule REJECTED — its premise (a clean low-end count gap) is not met; the largest ratio sits at the top end.
Overdispersed genes	1,670 (scFates GAM, data-driven).
PCA / n_pcs	n_pcs=30, precedent-bounded (Kurten 20 / Puram 40); Gate-1 call robust at 20/30/40.
Integration	Harmony(sample); theta swept 2/5/10/20(+50). theta=10 = best (max iLISI); none integrates.
Geometry	None — no trajectory fit (Gate 1 NO-GO). Manifold is a per-sample archipelago.
EMT-TFs shown	TWIST1 (26.8% expressing) · SNAI1 (6.0%) — two broadest-detected of a 10-TF panel; panel widened to FOXC/PRRX families; VIM excluded (a marker, not a TF).

VERDICT

The embryonic-program pattern does NOT reproduce in Dong 2020 prostate cancer: the malignant compartment does not form a shared, sample-integrated cell-state manifold. It fragments into per-patient islands — clonal/CNV heterogeneity plus a distinct small-cell NEPC pole (patient #6) **— so no trajectory is fit. Gate 1: NO-GO · Gates 2 & 3: NA.**